

Matchup Difference Analysis

Erika Nares

2026-05-31

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    4.0.0      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.2.0
## — Conflicts — tidyverse_conflicts() —
## * dplyr::filter() masks stats::filter()
## * dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(reticulate)
```

```
## Warning: package 'reticulate' was built under R version 4.4.3
```

```
combined_df <- read.csv("combined_df.csv")
```

```
from nba_api.stats.endpoints import leaguegamelog
```

```
def get_playoff_game_logs(season_year):
```

```
    logs = leaguegamelog.LeagueGameLog(
        season = season_year,
        season_type_all_star = "Playoffs",
        player_or_team_abbreviation = "T",
        timeout = 60
    )
```

```
    df_logs = logs.get_data_frames()[0]
```

```
    return df_logs
```

```

get_playoff_series_data <- function(season_year){

  df_logs <- py$get_playoff_game_logs(season_year)

  abbrev_lookup <- df_logs %>%
    select(Team_ABBREVIATION, Team_NAME) %>%
    distinct() %>%
    rename(
      OPP_ABBREVIATION = Team_ABBREVIATION,
      OPP_TEAM_NAME = Team_NAME
    )

  playoff_games <- df_logs %>%
    select(
      Team_NAME,
      Team_ABBREVIATION,
      MATCHUP,
      WL
    ) %>%
    mutate(
      OPP_ABBREVIATION = str_sub(MATCHUP, -3, -1),
      WIN = ifelse(WL == "W", 1, 0)
    ) %>%
    left_join(abbrev_lookup, by = "OPP_ABBREVIATION") %>%
    mutate(
      SERIES_ID = map2_chr(
        Team_NAME,
        OPP_TEAM_NAME,
        ~ paste(sort(c(.x, .y)), collapse = " vs ")
      )
    )

  team_series <- playoff_games %>%
    group_by(SERIES_ID, Team_NAME, OPP_TEAM_NAME) %>%
    summarize(
      SERIES_WINS = sum(WIN),
      .groups = "drop"
    )

  playoff_series <- team_series %>%
    group_by(SERIES_ID) %>%
    arrange(Team_NAME) %>%
    summarize(
      season = season_year,
      Team_A = first(Team_NAME),
      Team_B = last(Team_NAME),
      Team_A_WINS = SERIES_WINS[Team_NAME == Team_A],
      Team_B_WINS = SERIES_WINS[Team_NAME == Team_B],
      WINNER = ifelse(Team_A_WINS > Team_B_WINS, 1, 0),
      .groups = "drop"
    )

  return(playoff_series)
}

head(get_playoff_series_data("2014-15"))

```

```
## # A tibble: 6 × 7
##   SERIES_ID          season TEAM_A TEAM_B TEAM_A_WINS TEAM_B_WINS WINNER
##   <chr>          <chr>  <chr>  <chr>      <dbl>      <dbl>  <dbl>
## 1 Atlanta Hawks vs Brooklyn... 2014-... Atlan... Brook...      4          2      1
## 2 Atlanta Hawks vs Clevelan... 2014-... Atlan... Cleve...      0          4      0
## 3 Atlanta Hawks vs Washingt... 2014-... Atlan... Washi...      4          2      1
## 4 Boston Celtics vs Clevela... 2014-... Bosto... Cleve...      0          4      0
## 5 Chicago Bulls vs Clevelan... 2014-... Chica... Cleve...      2          4      0
## 6 Chicago Bulls vs Milwauke... 2014-... Chica... Milwa...
```

```
years <- c(
  "2015-16", "2016-17", "2017-18", "2018-19", "2019-20",
  "2020-21", "2021-22", "2022-23", "2023-24", "2024-25")

all_series <- map_dfr(years, function(year) {
  Sys.sleep(5)
  get_playoff_series_data(year)
})

head(all_series)
```

```
## # A tibble: 6 × 7
##   SERIES_ID          season TEAM_A TEAM_B TEAM_A_WINS TEAM_B_WINS WINNER
##   <chr>          <chr>  <chr>  <chr>      <dbl>      <dbl>  <dbl>
## 1 Atlanta Hawks vs Boston C... 2015-... Atlan... Bosto...      4          2      1
## 2 Atlanta Hawks vs Clevelan... 2015-... Atlan... Cleve...      0          4      0
## 3 Charlotte Hornets vs Miam... 2015-... Charl... Miami...      3          4      0
## 4 Cleveland Cavaliers vs De... 2015-... Cleve... Detro...      4          0      1
## 5 Cleveland Cavaliers vs Go... 2015-... Cleve... Golde...      4          3      1
## 6 Cleveland Cavaliers vs To... 2015-... Cleve... Toron...
```

```
write.csv(all_series, "all_series.csv", row.names = FALSE)
```

```
series_df <- all_series %>%
  left_join(
    combined_df,
    by = c(
      "TEAM_A" = "TEAM_NAME",
      "season" = "season"
    )
  )
head(series_df)
```

```
## # A tibble: 6 × 21
##   SERIES_ID      season TEAM_A TEAM_B TEAM_A_WINS TEAM_B_WINS WINNER TEAM_ID
##   <chr>          <chr>  <chr>  <chr>      <dbl>      <dbl>  <dbl>  <int>
## 1 Atlanta Hawks vs ... 2015-... Atlan... Bosto...      4        2      1 1.61e9
## 2 Atlanta Hawks vs ... 2015-... Atlan... Cleve...      0        4      0 1.61e9
## 3 Charlotte Hornets... 2015-... Charl... Miami...      3        4      0 1.61e9
## 4 Cleveland Cavalie... 2015-... Cleve... Detro...      4        0      1 1.61e9
## 5 Cleveland Cavalie... 2015-... Cleve... Golde...      4        3      1 1.61e9
## 6 Cleveland Cavalie... 2015-... Cleve... Toron...      4        2      1 1.61e9
## # i 13 more variables: W_PCT_REG <dbl>, OFF_RATING_REG <dbl>,
## #   DEF_RATING_REG <dbl>, NET_RATING_REG <dbl>, TS_PCT_REG <dbl>,
## #   EFG_PCT_REG <dbl>, OREB_PCT_REG <dbl>, TOV_PCT_REG <dbl>,
## #   PLUS_MINUS_REG <dbl>, THREE_PT_ATTEMPT_RATE_REG <dbl>, W_PCT_PO <dbl>,
## #   SOS_REG <dbl>, SRS_REG <dbl>
```

```
series_df <- series_df %>%
  left_join(
    combined_df,
    by = c(
      "TEAM_B" = "TEAM_NAME",
      "season" = "season"
    ),
    suffix = c("_A", "_B")
  )
head(series_df)
```

```
## # A tibble: 6 × 35
##   SERIES_ID      season TEAM_A TEAM_B TEAM_A_WINS TEAM_B_WINS WINNER TEAM_ID_A
##   <chr>          <chr>  <chr>  <chr>      <dbl>      <dbl>  <dbl>  <int>
## 1 Atlanta Hawks v... 2015-... Atlan... Bosto...      4        2      1 1.61e9
## 2 Atlanta Hawks v... 2015-... Atlan... Cleve...      0        4      0 1.61e9
## 3 Charlotte Horne... 2015-... Charl... Miami...      3        4      0 1.61e9
## 4 Cleveland Caval... 2015-... Cleve... Detro...      4        0      1 1.61e9
## 5 Cleveland Caval... 2015-... Cleve... Golde...      4        3      1 1.61e9
## 6 Cleveland Caval... 2015-... Cleve... Toron...      4        2      1 1.61e9
## # i 27 more variables: W_PCT_REG_A <dbl>, OFF_RATING_REG_A <dbl>,
## #   DEF_RATING_REG_A <dbl>, NET_RATING_REG_A <dbl>, TS_PCT_REG_A <dbl>,
## #   EFG_PCT_REG_A <dbl>, OREB_PCT_REG_A <dbl>, TOV_PCT_REG_A <dbl>,
## #   PLUS_MINUS_REG_A <dbl>, THREE_PT_ATTEMPT_RATE_REG_A <dbl>,
## #   W_PCT_PO_A <dbl>, SOS_REG_A <dbl>, SRS_REG_A <dbl>, TEAM_ID_B <int>,
## #   W_PCT_REG_B <dbl>, OFF_RATING_REG_B <dbl>, DEF_RATING_REG_B <dbl>,
## #   NET_RATING_REG_B <dbl>, TS_PCT_REG_B <dbl>, EFG_PCT_REG_B <dbl>, ...
```

```

series_df <- series_df %>%
  mutate(

    WIN_PCT_DIFF =
      W_PCT_REG_A -
      W_PCT_REG_B,

    NET_RATING_DIFF =
      NET_RATING_REG_A -
      NET_RATING_REG_B,

    TS_DIFF =
      TS_PCT_REG_A -
      TS_PCT_REG_B,

    THREE_PT_DIFF =
      THREE_PT_ATTEMPT_RATE_REG_A -
      THREE_PT_ATTEMPT_RATE_REG_B,

    OFF_RATING_DIFF =
      OFF_RATING_REG_A -
      OFF_RATING_REG_B,

    DEF_RATING_DIFF =
      DEF_RATING_REG_A -
      DEF_RATING_REG_B,

    EFG_DIFF =
      EFG_PCT_REG_A -
      EFG_PCT_REG_B,

    OREB_DIFF =
      OREB_PCT_REG_A -
      OREB_PCT_REG_B,

    TOV_DIFF =
      TOV_PCT_REG_A -
      TOV_PCT_REG_B,

    PLUS_MINUS_DIFF =
      PLUS_MINUS_REG_A -
      PLUS_MINUS_REG_B,

    SOS_DIFF =
      SOS_REG_A -
      SOS_REG_B,

    SRS_DIFF =
      SRS_REG_A -
      SRS_REG_B

  )
head(series_df)

```

```
## # A tibble: 6 × 47
##   SERIES_ID      season TEAM_A TEAM_B TEAM_A_WINS TEAM_B_WINS WINNER TEAM_ID_A
##   <chr>         <chr>  <chr>  <chr>         <dbl>         <dbl>  <dbl>         <int>
## 1 Atlanta Hawks v... 2015-... Atlan... Bosto...         4           2         1         1.61e9
## 2 Atlanta Hawks v... 2015-... Atlan... Cleve...         0           4         0         1.61e9
## 3 Charlotte Horne... 2015-... Charl... Miami...         3           4         0         1.61e9
## 4 Cleveland Caval... 2015-... Cleve... Detro...         4           0         1         1.61e9
## 5 Cleveland Caval... 2015-... Cleve... Golde...         4           3         1         1.61e9
## 6 Cleveland Caval... 2015-... Cleve... Toron...         4           2         1         1.61e9
## # i 39 more variables: W_PCT_REG_A <dbl>, OFF_RATING_REG_A <dbl>,
## #   DEF_RATING_REG_A <dbl>, NET_RATING_REG_A <dbl>, TS_PCT_REG_A <dbl>,
## #   EFG_PCT_REG_A <dbl>, OREB_PCT_REG_A <dbl>, TOV_PCT_REG_A <dbl>,
## #   PLUS_MINUS_REG_A <dbl>, THREE_PT_ATTEMPT_RATE_REG_A <dbl>,
## #   W_PCT_PO_A <dbl>, SOS_REG_A <dbl>, SRS_REG_A <dbl>, TEAM_ID_B <int>,
## #   W_PCT_REG_B <dbl>, OFF_RATING_REG_B <dbl>, DEF_RATING_REG_B <dbl>,
## #   NET_RATING_REG_B <dbl>, TS_PCT_REG_B <dbl>, EFG_PCT_REG_B <dbl>, ...
```

```
series_df <- series_df %>% select(SERIES_ID, season, TEAM_A, TEAM_B, WINNER, TEAM_ID_A, TEAM_ID_
B, WIN_PCT_DIFF,
                                NET_RATING_DIFF, TS_DIFF, THREE_PT_DIFF, OFF_RATING_DIFF, DEF_R
ATING_DIFF,
                                EFG_DIFF, OREB_DIFF, TOV_DIFF, PLUS_MINUS_DIFF, SOS_DIFF, SRS_D
IFF)
```

```
head(series_df)
```

```
## # A tibble: 6 × 19
##   SERIES_ID      season TEAM_A TEAM_B WINNER TEAM_ID_A TEAM_ID_B WIN_PCT_DIFF
##   <chr>         <chr>  <chr>  <chr>         <dbl>         <int>         <int>         <dbl>
## 1 Atlanta Hawks vs... 2015-... Atlan... Bosto...         1         1.61e9         1.61e9         0
## 2 Atlanta Hawks vs... 2015-... Atlan... Cleve...         0         1.61e9         1.61e9        -0.11
## 3 Charlotte Hornet... 2015-... Charl... Miami...         0         1.61e9         1.61e9         0
## 4 Cleveland Cavali... 2015-... Cleve... Detro...         1         1.61e9         1.61e9         0.158
## 5 Cleveland Cavali... 2015-... Cleve... Golde...         1         1.61e9         1.61e9        -0.195
## 6 Cleveland Cavali... 2015-... Cleve... Toron...         1         1.61e9         1.61e9         0.0120
## # i 11 more variables: NET_RATING_DIFF <dbl>, TS_DIFF <dbl>,
## #   THREE_PT_DIFF <dbl>, OFF_RATING_DIFF <dbl>, DEF_RATING_DIFF <dbl>,
## #   EFG_DIFF <dbl>, OREB_DIFF <dbl>, TOV_DIFF <dbl>, PLUS_MINUS_DIFF <dbl>,
## #   SOS_DIFF <dbl>, SRS_DIFF <dbl>
```

```
write.csv(series_df, "series_df.csv", row.names = FALSE)
```

```

difference_cors <- series_df %>% group_by(season)%>%
  summarize(
    win_pct_diff_cor = cor(WIN_PCT_DIFF, WINNER),
    net_rating_diff_cor = cor(NET_RATING_DIFF, WINNER),
    plus_minus_diff_cor = cor(PLUS_MINUS_DIFF, WINNER),
    off_rating_diff_cor = cor(OFF_RATING_DIFF, WINNER),
    def_rating_diff_cor = cor(DEF_RATING_DIFF, WINNER),
    ts_diff_cor = cor(TS_DIFF, WINNER),
    efg_diff_cor = cor(EFG_DIFF, WINNER),
    three_pt_diff_cor = cor(THREE_PT_DIFF, WINNER),
    tov_diff_cor = cor(TOV_DIFF, WINNER),
    oreb_diff_cor = cor(OREB_DIFF, WINNER),
    sos_diff_cor = cor(SOS_DIFF, WINNER),
    srs_diff_cor = cor(SRS_DIFF, WINNER)
  )

head(difference_cors)

```

```

## # A tibble: 6 × 13
##   season win_pct_diff_cor net_rating_diff_cor plus_minus_diff_cor
##   <chr>      <dbl>          <dbl>          <dbl>
## 1 2015-16      0.504            0.527            0.526
## 2 2016-17      0.773            0.743            0.744
## 3 2017-18      0.441            0.360            0.361
## 4 2018-19      0.711            0.621            0.622
## 5 2019-20      0.469            0.367            0.363
## 6 2020-21      0.476            0.552            0.550
## # i 9 more variables: off_rating_diff_cor <dbl>, def_rating_diff_cor <dbl>,
## #   ts_diff_cor <dbl>, efg_diff_cor <dbl>, three_pt_diff_cor <dbl>,
## #   tov_diff_cor <dbl>, oreb_diff_cor <dbl>, sos_diff_cor <dbl>,
## #   srs_diff_cor <dbl>

```

```
difference_correlations <- tibble(
```

```
  Variable = c(
    "WIN_PCT_DIFF",
    "NET_RATING_DIFF",
    "TS_DIFF",
    "THREE_PT_DIFF",
    "OFF_RATING_DIFF",
    "DEF_RATING_DIFF",
    "EFG_DIFF",
    "OREB_DIFF",
    "TOV_DIFF",
    "PLUS_MINUS_DIFF",
    "SOS_DIFF",
    "SRS_DIFF"
  ),
```

```
  Correlation = c(
    cor(series_df$WIN_PCT_DIFF,
        series_df$WINNER),

    cor(series_df$NET_RATING_DIFF,
        series_df$WINNER),

    cor(series_df$TS_DIFF,
        series_df$WINNER),

    cor(series_df$THREE_PT_DIFF,
        series_df$WINNER),

    cor(series_df$OFF_RATING_DIFF,
        series_df$WINNER),

    cor(series_df$DEF_RATING_DIFF,
        series_df$WINNER),

    cor(series_df$EFG_DIFF,
        series_df$WINNER),

    cor(series_df$OREB_DIFF,
        series_df$WINNER),

    cor(series_df$TOV_DIFF,
        series_df$WINNER),

    cor(series_df$PLUS_MINUS_DIFF,
        series_df$WINNER),

    cor(series_df$SOS_DIFF,
        series_df$WINNER),

    cor(series_df$SRS_DIFF,
        series_df$WINNER)
  )
)
```



```
difference_correlations %>%  
  arrange(desc(abs(Correlation)))
```

```
## # A tibble: 12 × 2  
##   Variable      Correlation  
##   <chr>         <dbl>  
## 1 WIN_PCT_DIFF      0.504  
## 2 EFG_DIFF          0.489  
## 3 TS_DIFF           0.485  
## 4 NET_RATING_DIFF   0.478  
## 5 PLUS_MINUS_DIFF   0.477  
## 6 SRS_DIFF           0.477  
## 7 OFF_RATING_DIFF   0.457  
## 8 SOS_DIFF          -0.310  
## 9 DEF_RATING_DIFF   -0.219  
## 10 THREE_PT_DIFF    0.210  
## 11 OREB_DIFF         -0.0459  
## 12 TOV_DIFF          0.0148
```

```
diff_predictors <- series_df %>%  
  select(  
    WIN_PCT_DIFF,  
    NET_RATING_DIFF,  
    TS_DIFF,  
    THREE_PT_DIFF,  
    OFF_RATING_DIFF,  
    DEF_RATING_DIFF,  
    EFG_DIFF,  
    OREB_DIFF,  
    TOV_DIFF,  
    PLUS_MINUS_DIFF,  
    SOS_DIFF,  
    SRS_DIFF  
  )
```

```
round(  
  cor(  
    diff_predictors,  
    use = "complete.obs"  
  ),  
  2  
)
```

##	WIN_PCT_DIFF	NET_RATING_DIFF	TS_DIFF	THREE_PT_DIFF
## WIN_PCT_DIFF	1.00	0.94	0.64	0.30
## NET_RATING_DIFF	0.94	1.00	0.61	0.32
## TS_DIFF	0.64	0.61	1.00	0.38
## THREE_PT_DIFF	0.30	0.32	0.38	1.00
## OFF_RATING_DIFF	0.69	0.70	0.77	0.37
## DEF_RATING_DIFF	-0.64	-0.71	-0.10	-0.08
## EFG_DIFF	0.68	0.65	0.96	0.37
## OREB_DIFF	-0.10	0.00	-0.36	-0.24
## TOV_DIFF	-0.15	-0.11	0.09	-0.02
## PLUS_MINUS_DIFF	0.94	1.00	0.61	0.32
## SOS_DIFF	-0.70	-0.64	-0.49	-0.23
## SRS_DIFF	0.94	1.00	0.61	0.32

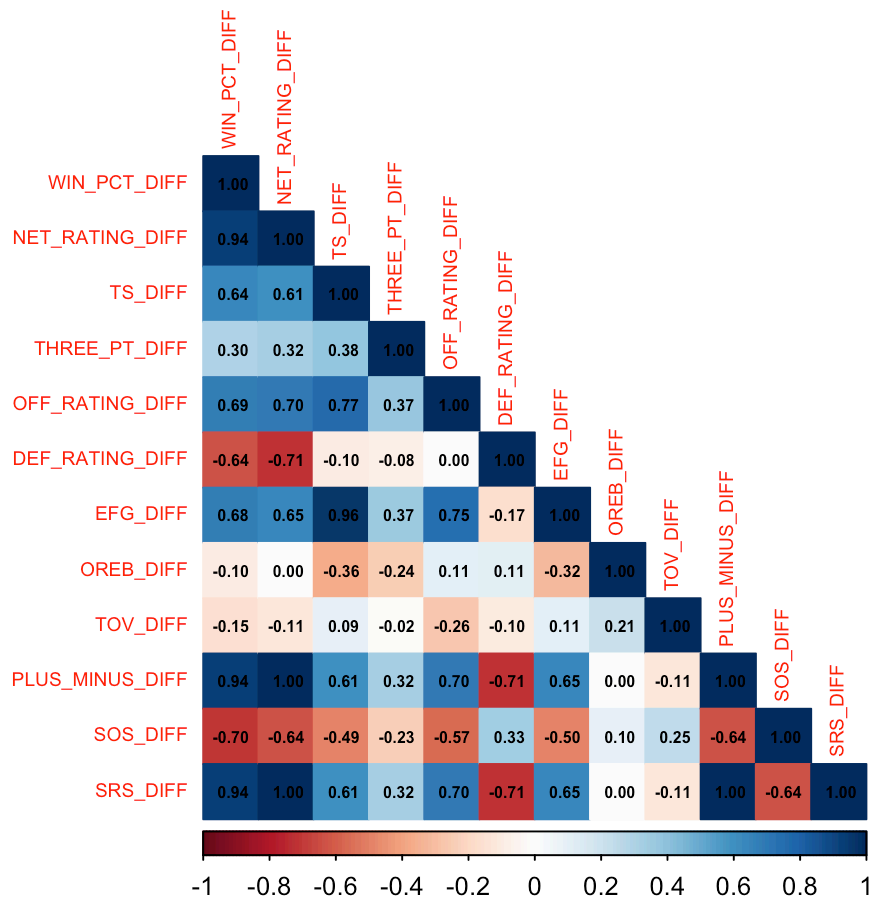
##	OFF_RATING_DIFF	DEF_RATING_DIFF	EFG_DIFF	OREB_DIFF	TOV_DIFF
## WIN_PCT_DIFF	0.69	-0.64	0.68	-0.10	-0.15
## NET_RATING_DIFF	0.70	-0.71	0.65	0.00	-0.11
## TS_DIFF	0.77	-0.10	0.96	-0.36	0.09
## THREE_PT_DIFF	0.37	-0.08	0.37	-0.24	-0.02
## OFF_RATING_DIFF	1.00	0.00	0.75	0.11	-0.26
## DEF_RATING_DIFF	0.00	1.00	-0.17	0.11	-0.10
## EFG_DIFF	0.75	-0.17	1.00	-0.32	0.11
## OREB_DIFF	0.11	0.11	-0.32	1.00	0.21
## TOV_DIFF	-0.26	-0.10	0.11	0.21	1.00
## PLUS_MINUS_DIFF	0.70	-0.71	0.65	0.00	-0.11
## SOS_DIFF	-0.57	0.33	-0.50	0.10	0.25
## SRS_DIFF	0.70	-0.71	0.65	0.00	-0.11

##	PLUS_MINUS_DIFF	SOS_DIFF	SRS_DIFF
## WIN_PCT_DIFF	0.94	-0.70	0.94
## NET_RATING_DIFF	1.00	-0.64	1.00
## TS_DIFF	0.61	-0.49	0.61
## THREE_PT_DIFF	0.32	-0.23	0.32
## OFF_RATING_DIFF	0.70	-0.57	0.70
## DEF_RATING_DIFF	-0.71	0.33	-0.71
## EFG_DIFF	0.65	-0.50	0.65
## OREB_DIFF	0.00	0.10	0.00
## TOV_DIFF	-0.11	0.25	-0.11
## PLUS_MINUS_DIFF	1.00	-0.64	1.00
## SOS_DIFF	-0.64	1.00	-0.64
## SRS_DIFF	1.00	-0.64	1.00

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
corrplot(
  cor(diff_predictors, use = "complete.obs"),
  method = "color",
  type = "lower",
  addCoef.col = "black",
  number.cex = 0.5,
  tl.cex = 0.6
)
```



- 1. Win percentage difference
- 2. TS Difference
- 3. Three Point Difference